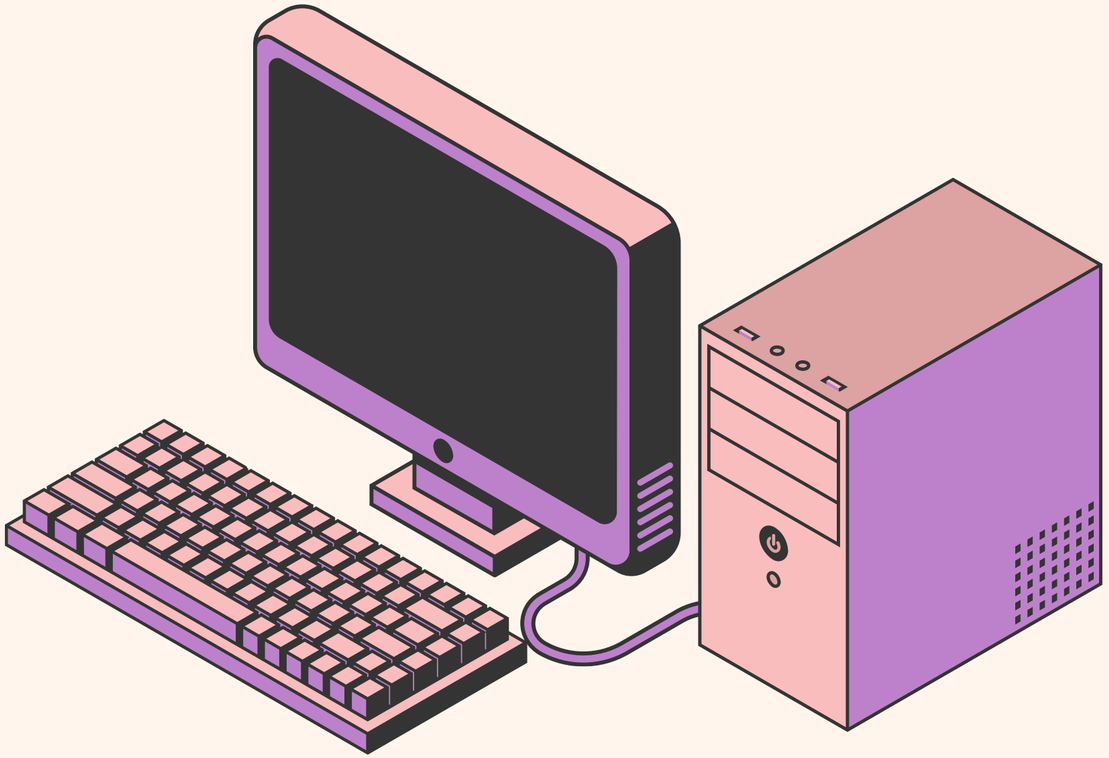
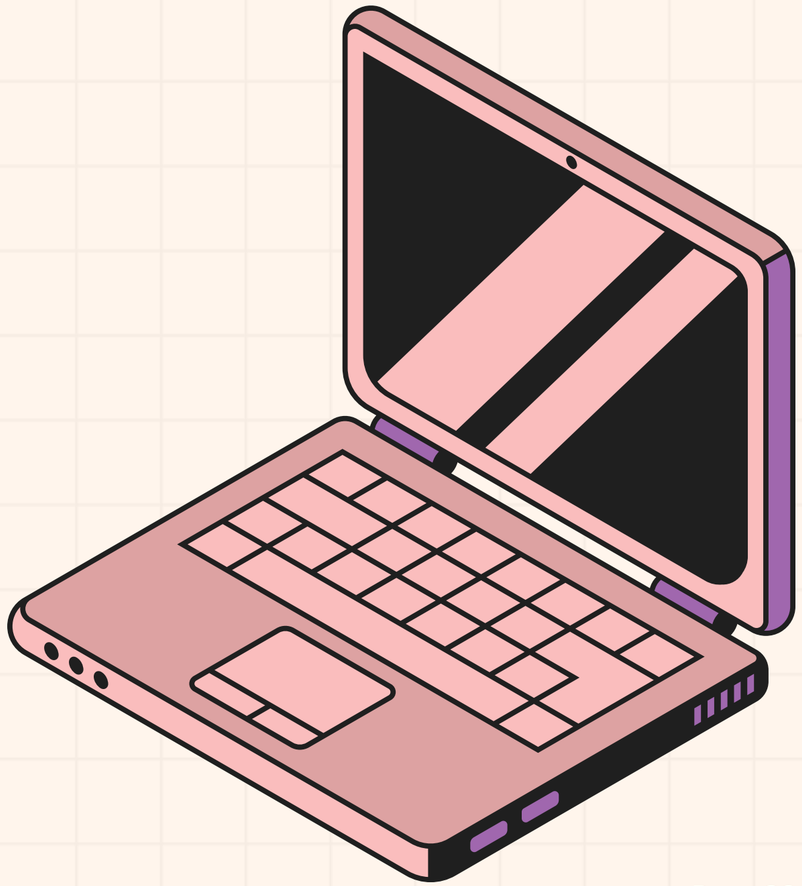
FORMATTING
& INSTALLATION

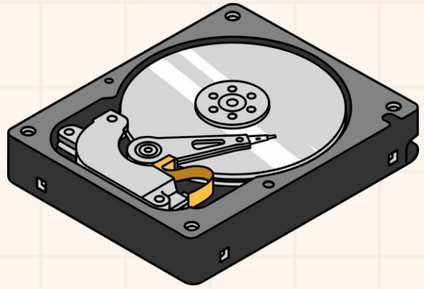


INTRODUCTION

Formatting a hard disk drive (HDD) and installing an operating system are essential steps in preparing a computer for use. Formatting organizes the storage device so data can be stored and accessed properly. This study explains the process of HDD formatting, the types of formatting, and the importance of file systems. It also discusses how an operating system is installed and why it is necessary for a computer to function efficiently.

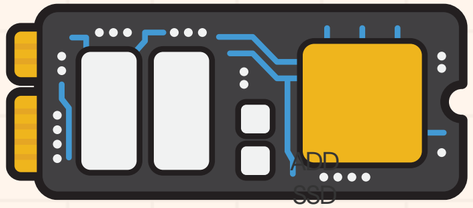


HARDWARE COMPONENTS



HARD DISK DRIVE (HDD)

A Hard Disk Drive (HDD) is a storage device used to save the operating system, software, and user files on a computer. It stores data permanently and allows the system to access information even after the computer is turned off.



SOLID STATE DRIVE (SSD)

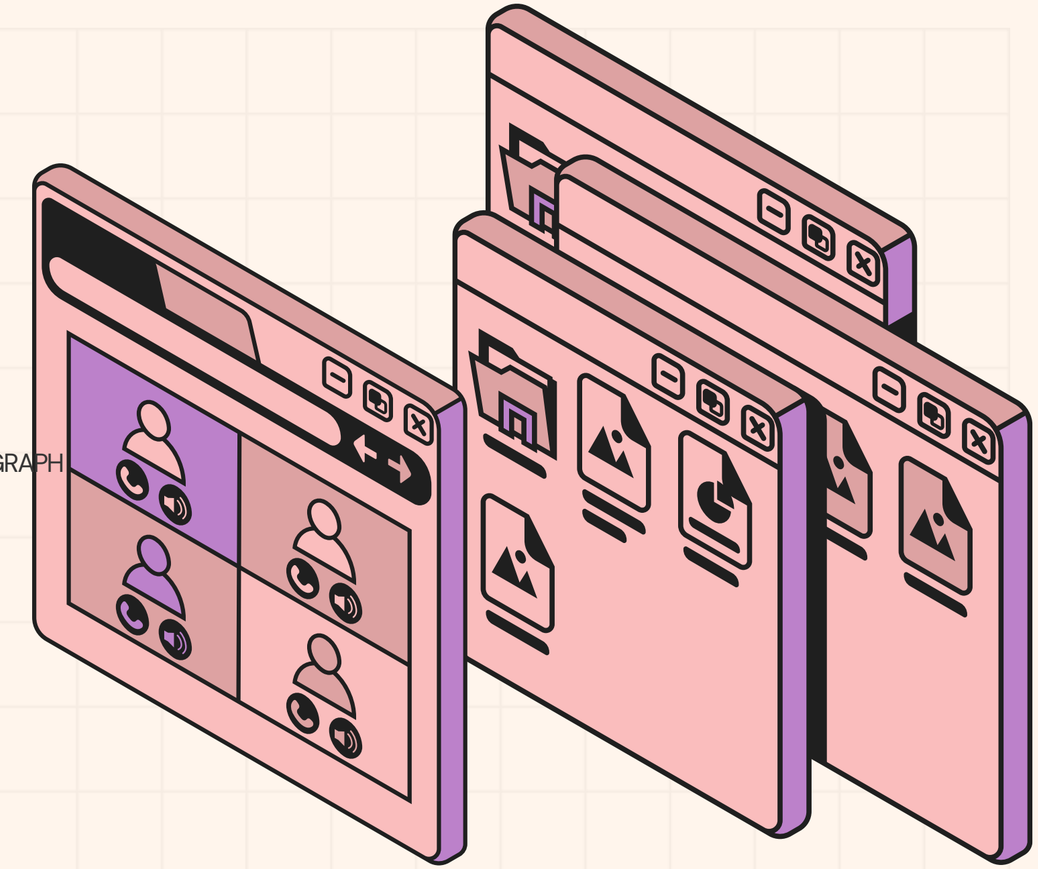
A Solid State Drive (SSD) is a fast, reliable storage device that uses flash memory instead of moving parts, allowing quicker boot times, faster file transfers, better durability, and lower power use than traditional hard disk drives (HDDs).

BOOTABLE USB

A bootable USB is a portable storage device that contains installation files for an operating system. It allows a computer to start up and install or repair an operating system without using the internal hard drive.

TYPES OF FORMATTING

YOUR PARAGRAPH TEXT

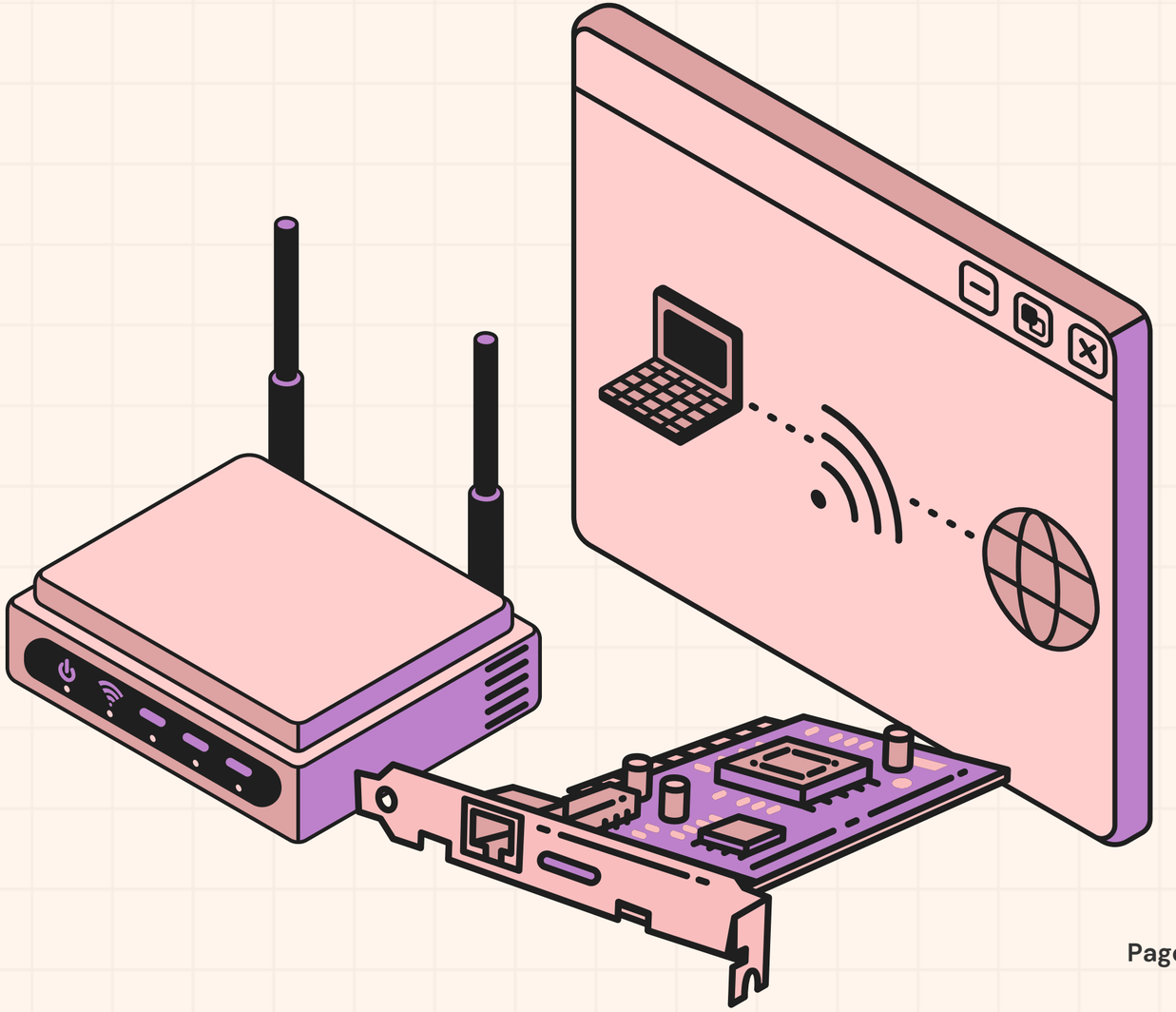


LOW-LEVEL FORMATTING

This type of formatting is performed by the manufacturer and prepares the physical structure of the hard disk. It is not usually done by regular users.

HIGH-LEVEL FORMATTING

High-level formatting creates a file system (such as NTFS or FAT32) on the hard disk so the operating system can store and manage data properly. This is the most common type of formatting done by users.

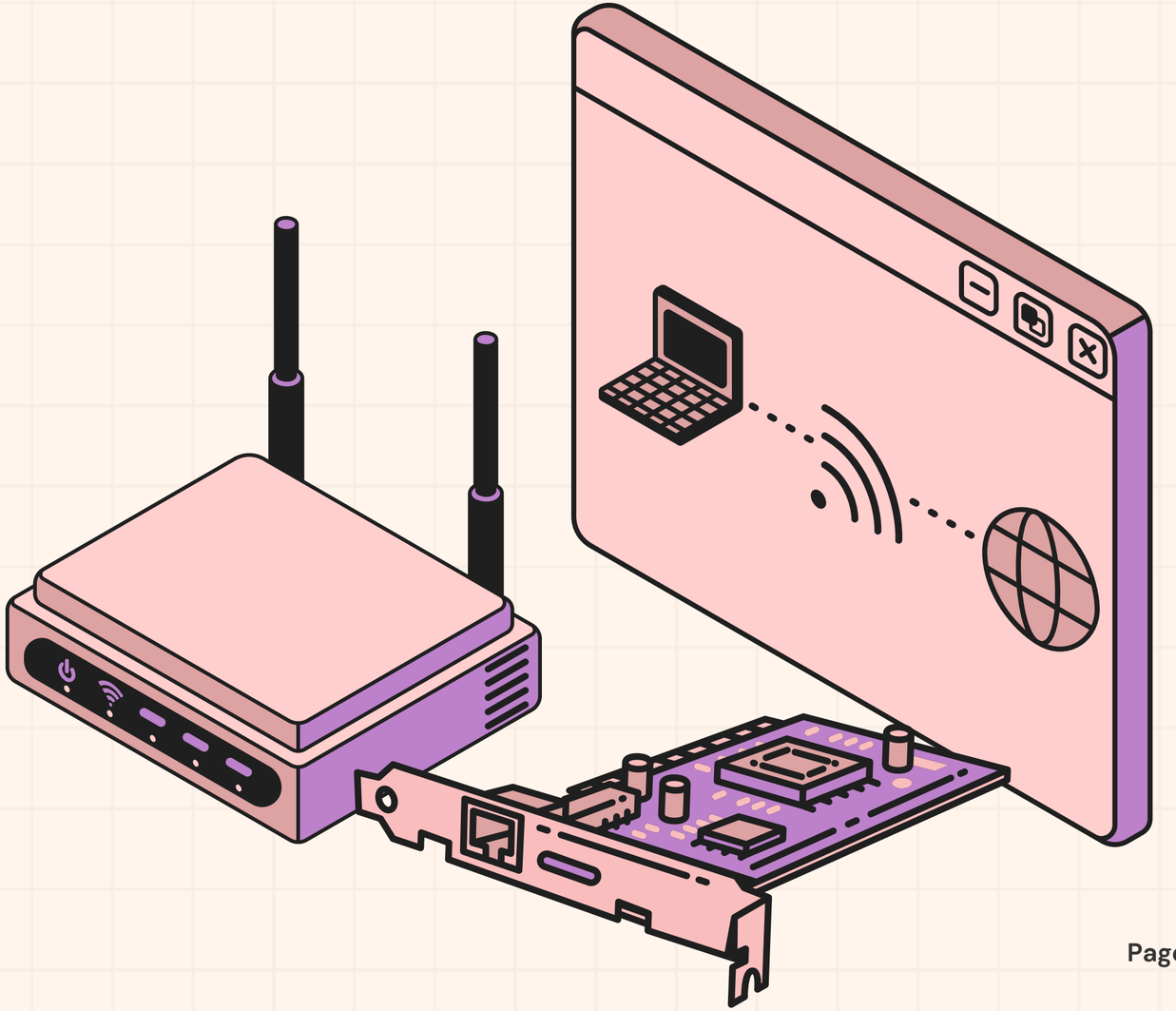


QUICK FORMATTING

When you perform a quick format, the operating system removes the file system structure—such as the file table and directory listings so the device appears empty. However, the actual data remains physically on the storage until it gets overwritten by new files

FULL FORMATTING

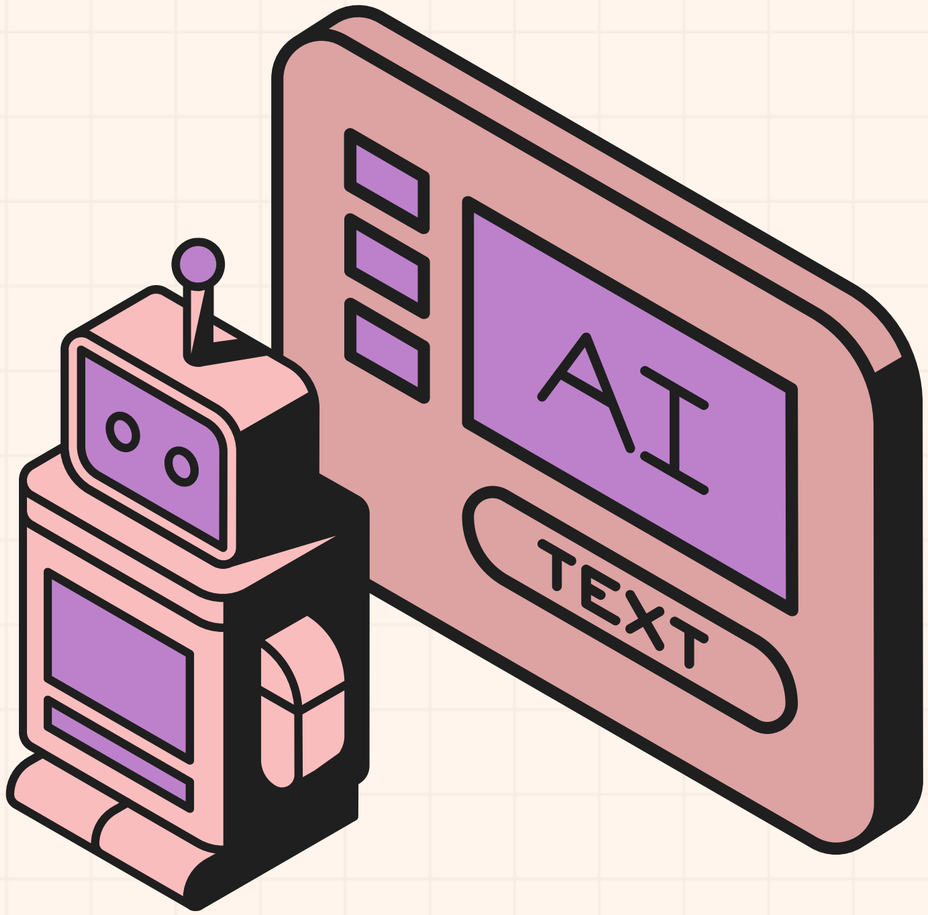
Full formatting completely erases all data on the hard disk and checks for disk errors. It takes longer but is more thorough and secure.



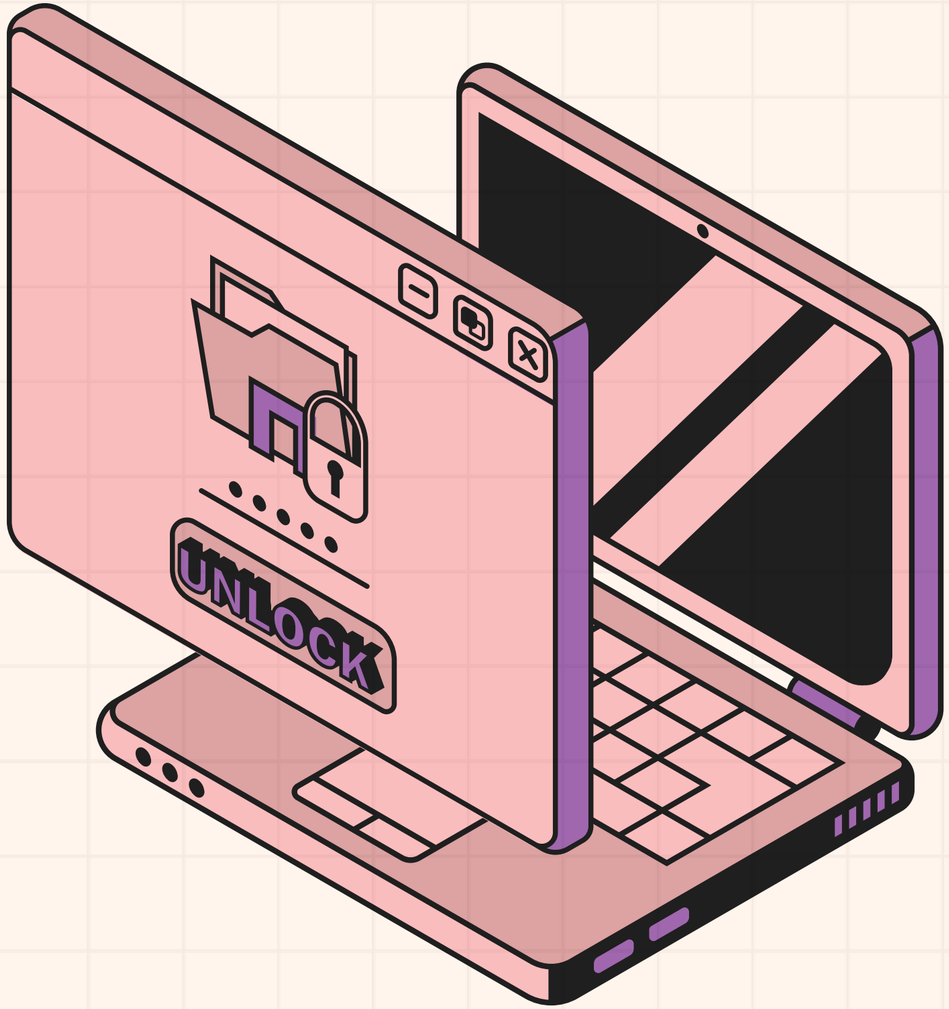
TOOLS NEEDED

- Bootable USB/DVD with OS installer
- BIOS/UEFI access
- Backup of important data
- Stable power supply

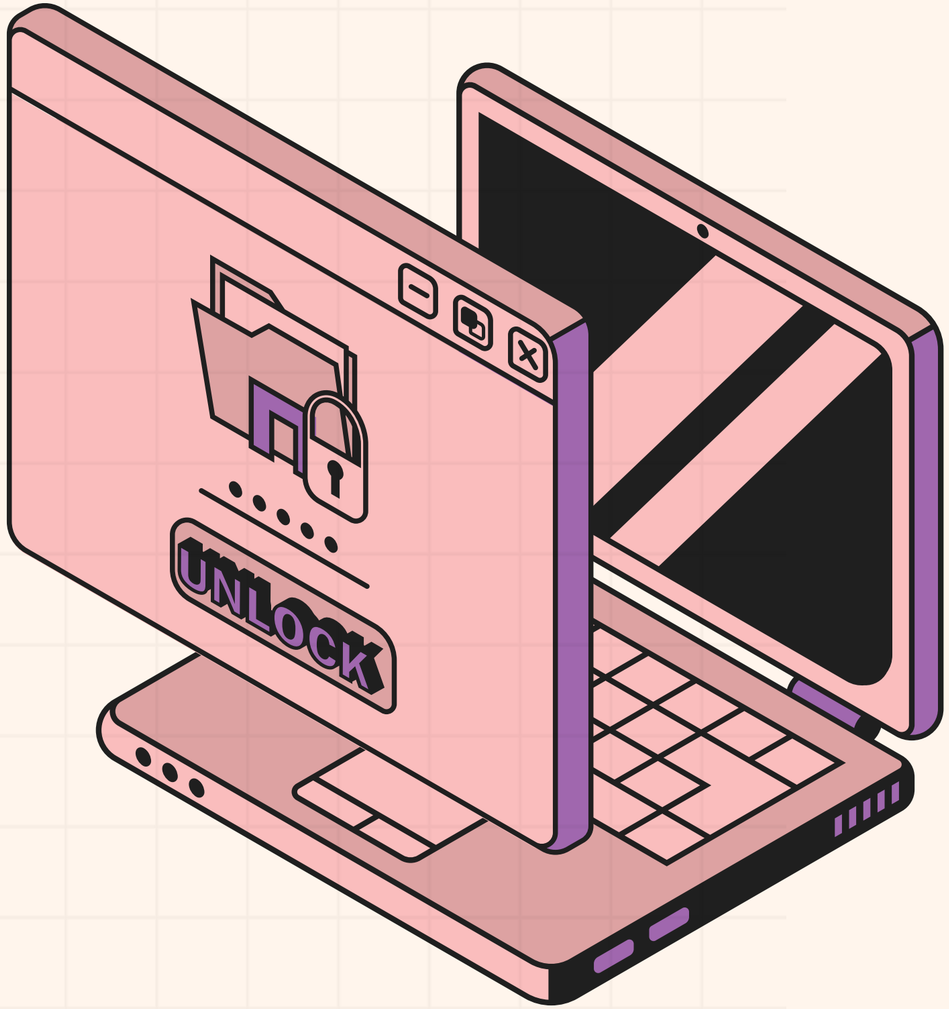
PROPER TOOLS PREVENT DATA LOSS AND INSTALLATION FAILURE.



FORMATTING/INSTALLATION OF OPERATING SYSTEM (OS)



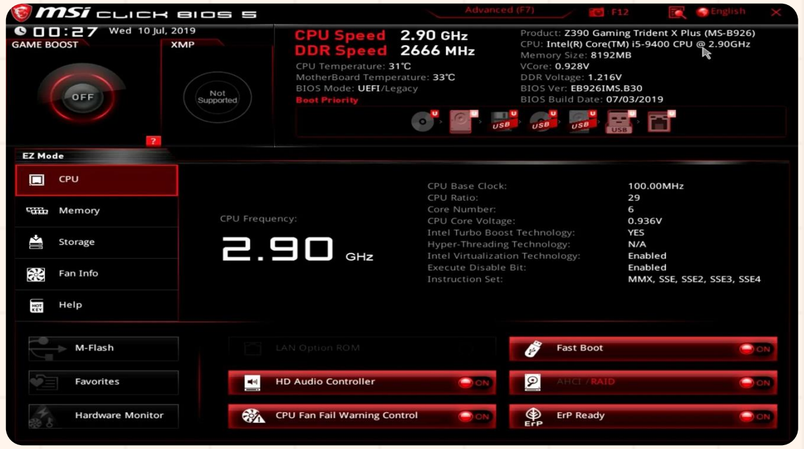
FORMATTING



STEP 1:

boot from installation media

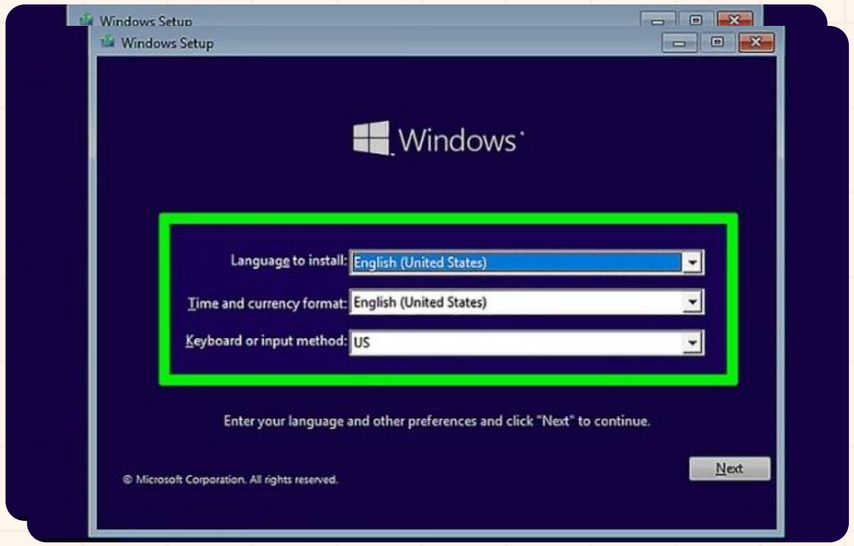
- **Insert the Bootable USB:** Plug in the portable storage device containing the OS installation files
- **Enter BIOS/UEFI:** Restart the computer and press the specific key (usually F2, F12, or DEL depending on the model of motherboard) to access the system setup.
- **Change Boot Priority**: Set the Bootable USB as the primary boot device so the computer starts from the installer rather than the internal hard drive.

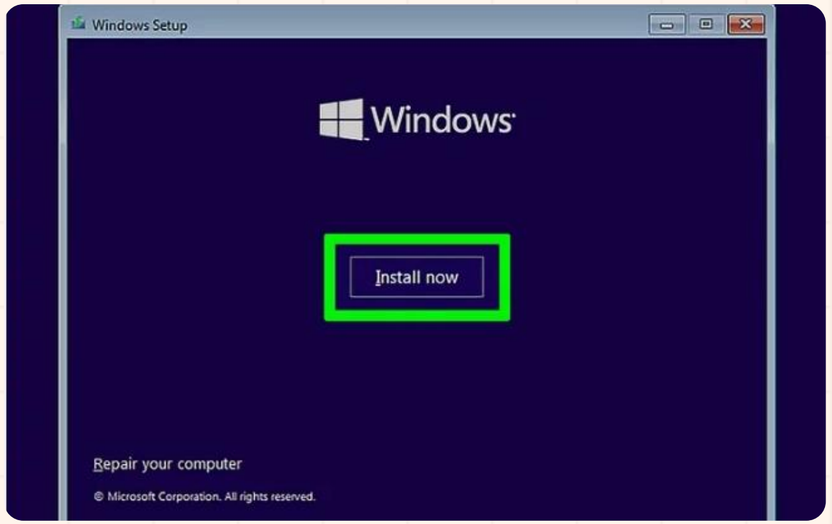


STEP 2:

begin the setup

- **Language and Region:** Select your preferred language, time zone, and keyboard layout when the installer screen appears.
- **Initiate Setup:** Proceed to the operating system setup

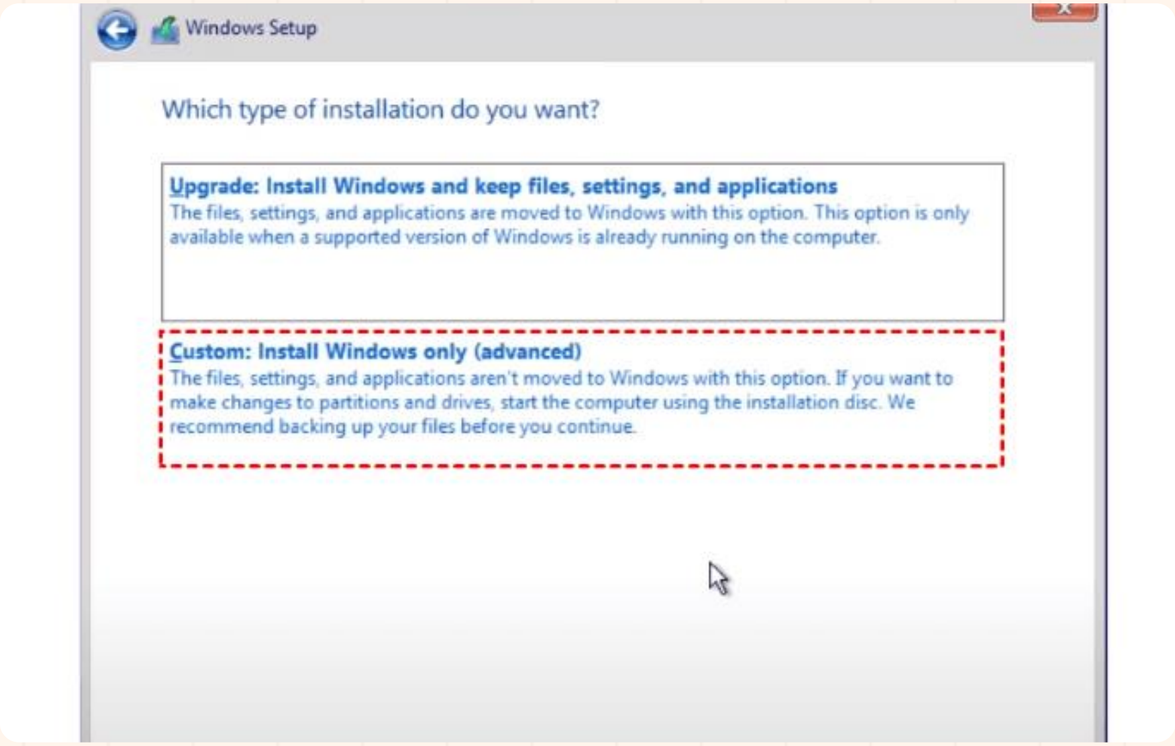




STEP 3:

select installation type

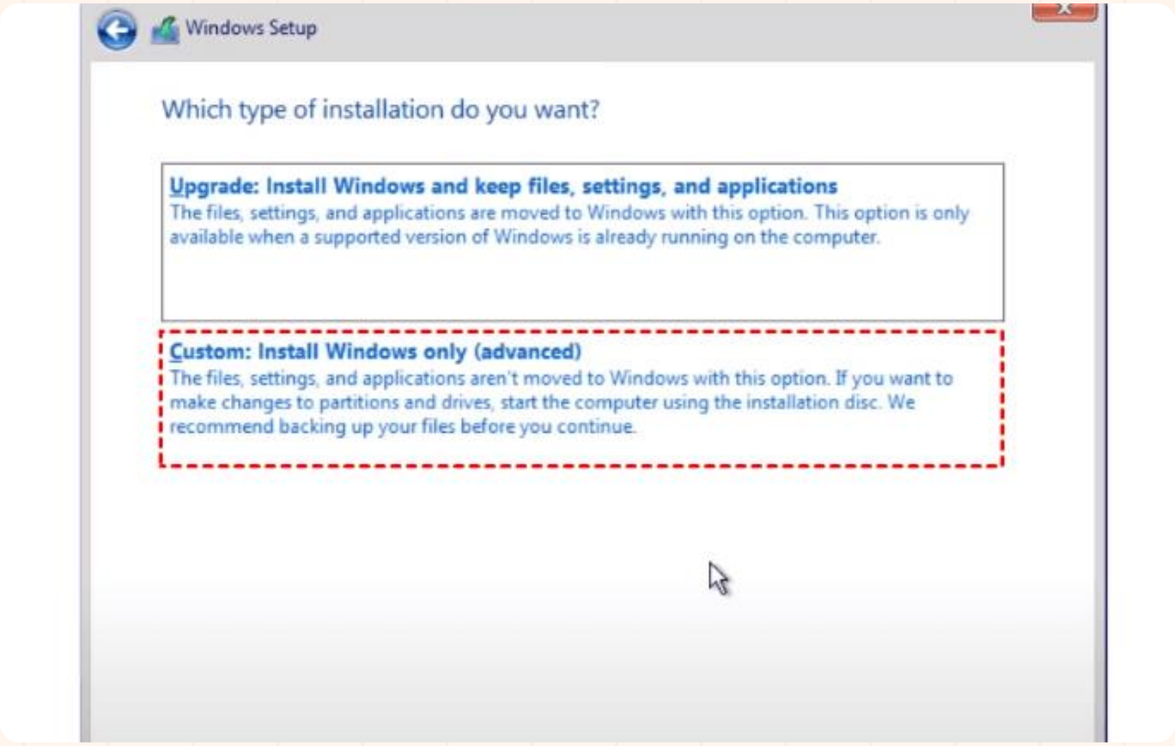
- **Custom Installation:** Since you have already formatted your drive or are using a "new or non-binary HDD", choose Custom: Install Windows only (advanced)



STEP 4:

Finalizing the Install

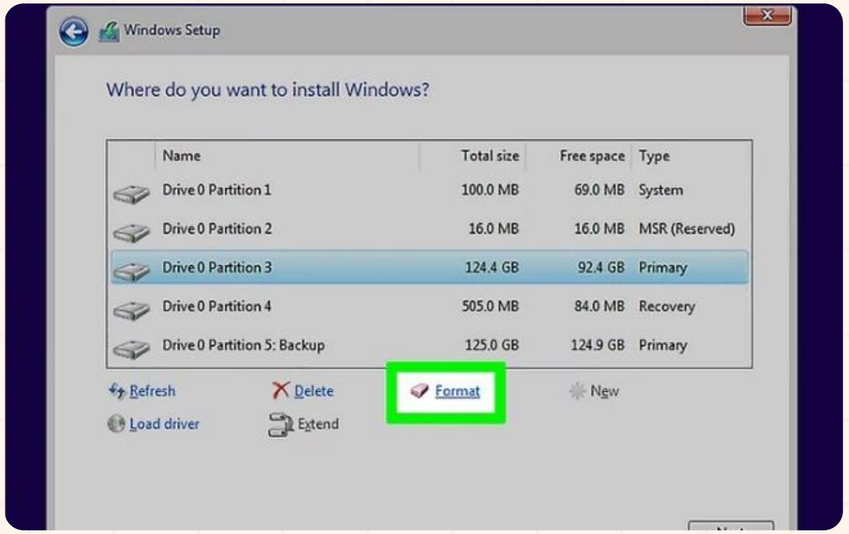
- **Installation Sequence:** The computer will begin copying files, installing features, and applying updates. The system may restart several times during this phase; ensure you have a stable power supply to prevent failure.
- **User Setup:** Once finished, follow the on-screen prompts to create a user account, set a password, and adjust privacy settings.

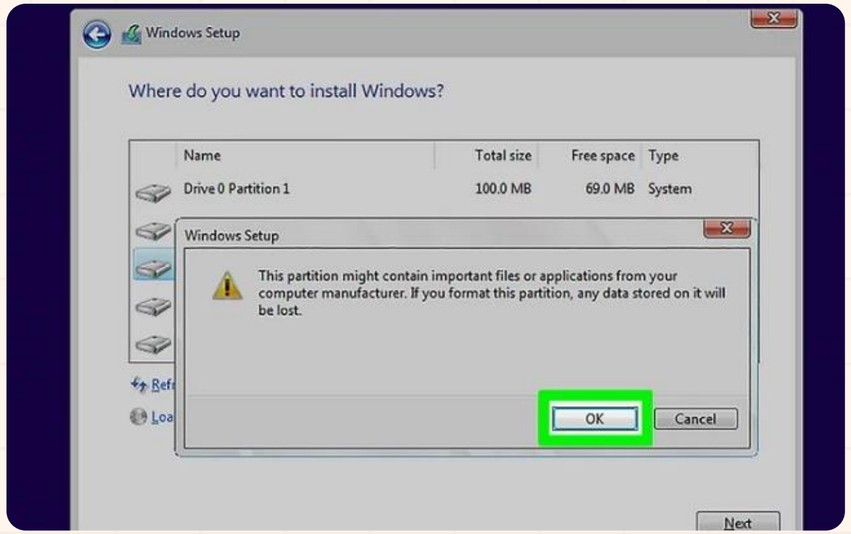


STEP 5:

Select the primary partition

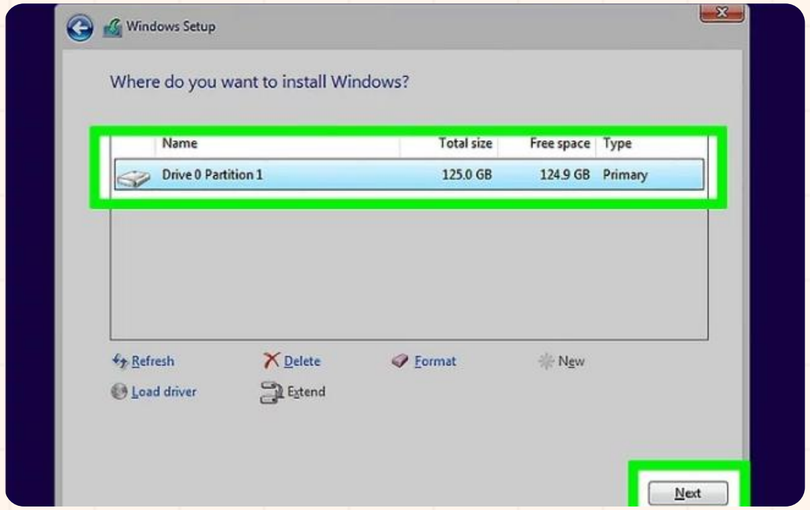
- This is the partition on which Windows was previously installed. It's the one that's labeled "Primary."
- Click OK on the warning message. This message just warns you that any data on the drive or partition will be lost if you continue. Once selected, the selected drive or partition will be formatted.



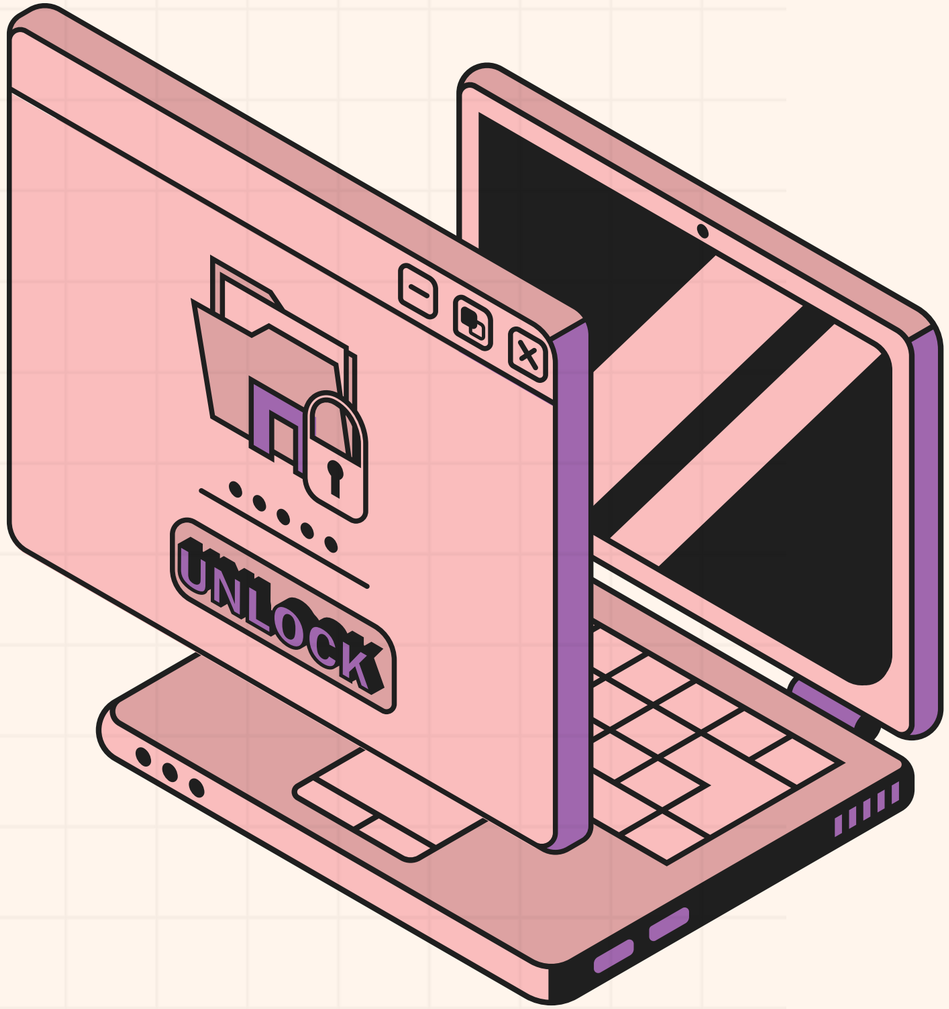


STEP 6:
FINALIZATION

- Now that you've formatted the drive, you can follow the on-screen instructions to perform a clean Windows installation. After Windows is installed, you'll be able to sign in with your Microsoft account (or create a new local account) and set up your computer like new.



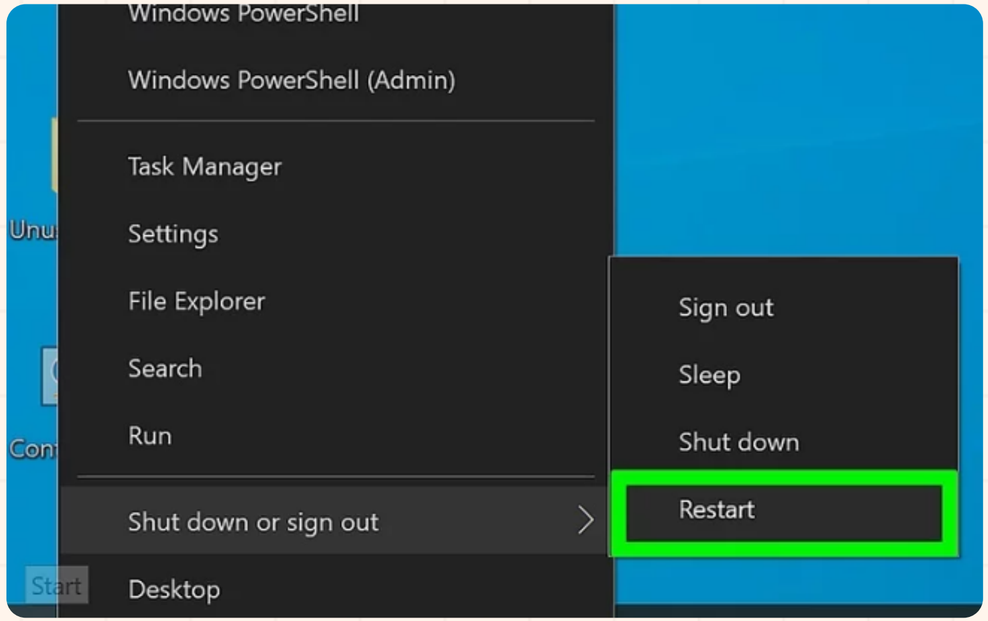
FORMATTING



STEP 1:

BOOT THE COMPUTER

- If the PC is already powered on, restart it using the normal restart procedures of the operating system that is installed on the PC. If it is not yet powered on, press the power button to boot it up.



STEP 2:

ENTER BOOT MENU/BIOS

- Enter the Boot menu. Most newer computers have a boot menu you can enter while booting up. The way you enter it is different, depending on your motherboard. The most common way to enter the Boot menu is to press F10, F11, F12, F2, or Esc as the PC boots up. This Boot menu will display a list of drives you can boot from

STEP 3:

select usb drive

- This is the drive that has the Windows Installation Media. This will boot the PC from the Windows Installation Media. Once the computer boots from the USB drive, press any key to start the Windows installation process.

STEP 4:

select preference

- Use the drop-down menus to first select your language. Then select your country or region, and your keyboard input method. Then click Next

STEP 5:

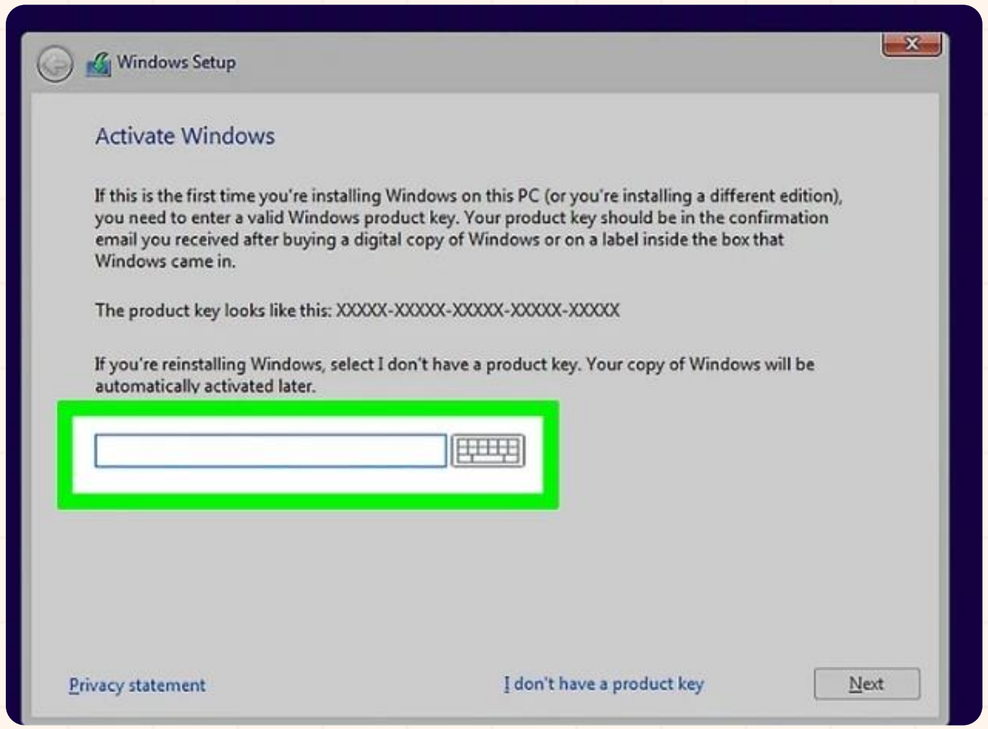
installation

- Click Install Now. It's the button in the center of the screen.

STEP 6:

PRODUCT KEY

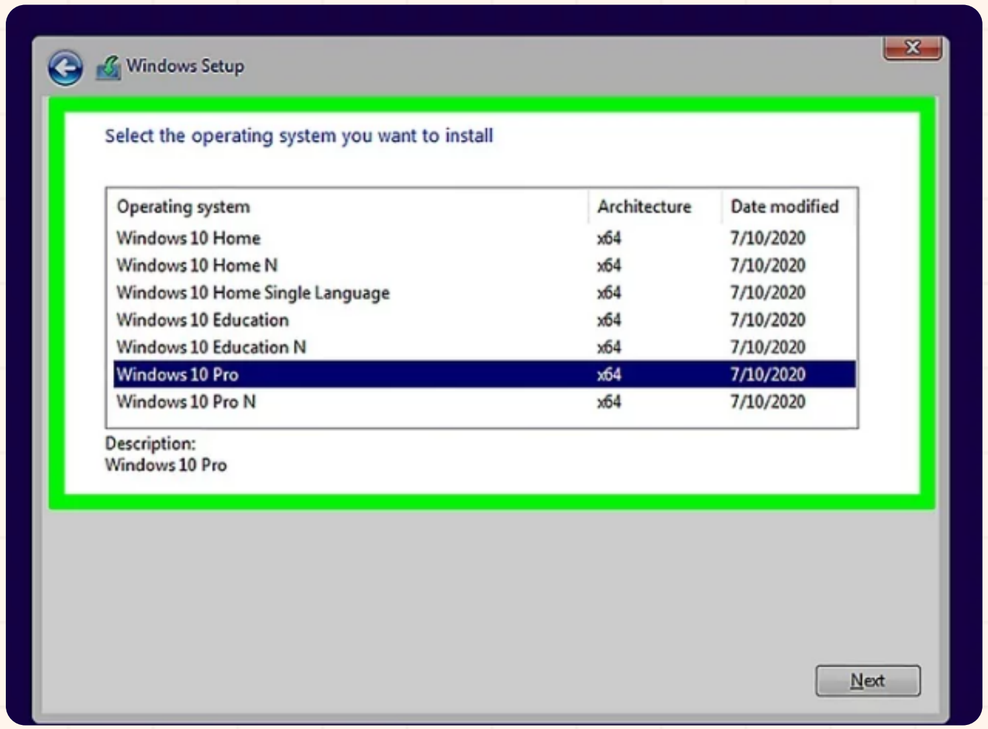
- If you've already purchased Windows, you can find your license key in your confirmation email. If you have not purchased Windows, click I don't have a product key. You'll need to purchase Windows and activate it later on.



STEP 7:

SELECT EDITION

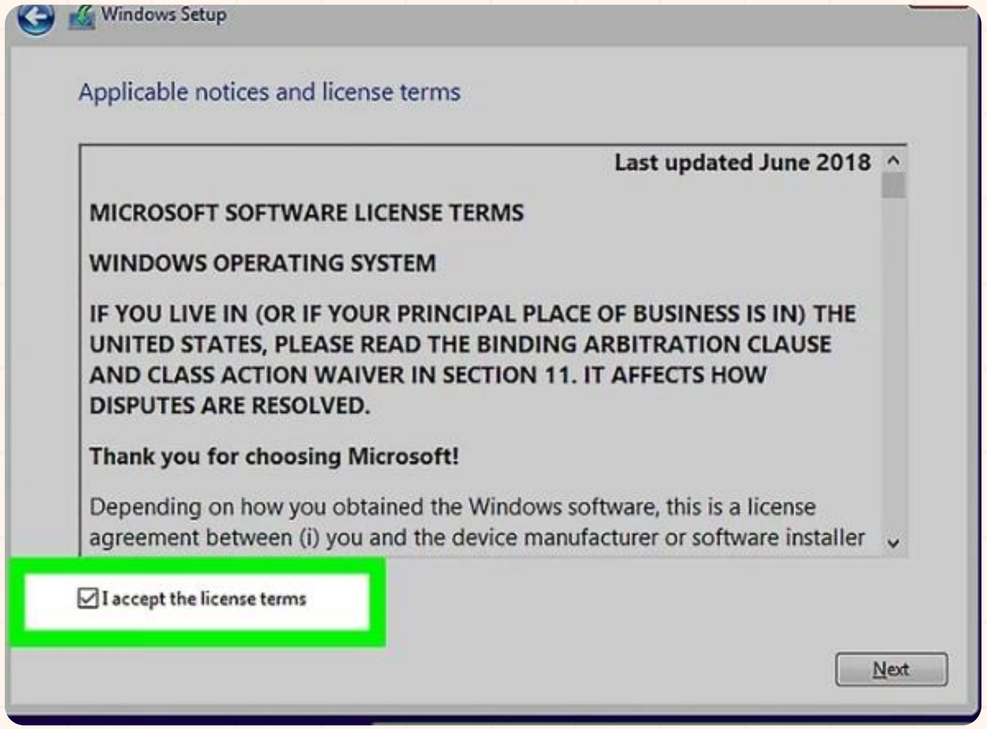
- If you've already purchased Windows, check which version of Windows you purchased and make sure you click the right version. If you have not yet purchased Windows, check which version you want to install. Different versions of Windows have a different price.



STEP 8:

CLICK THE TEXT BOX

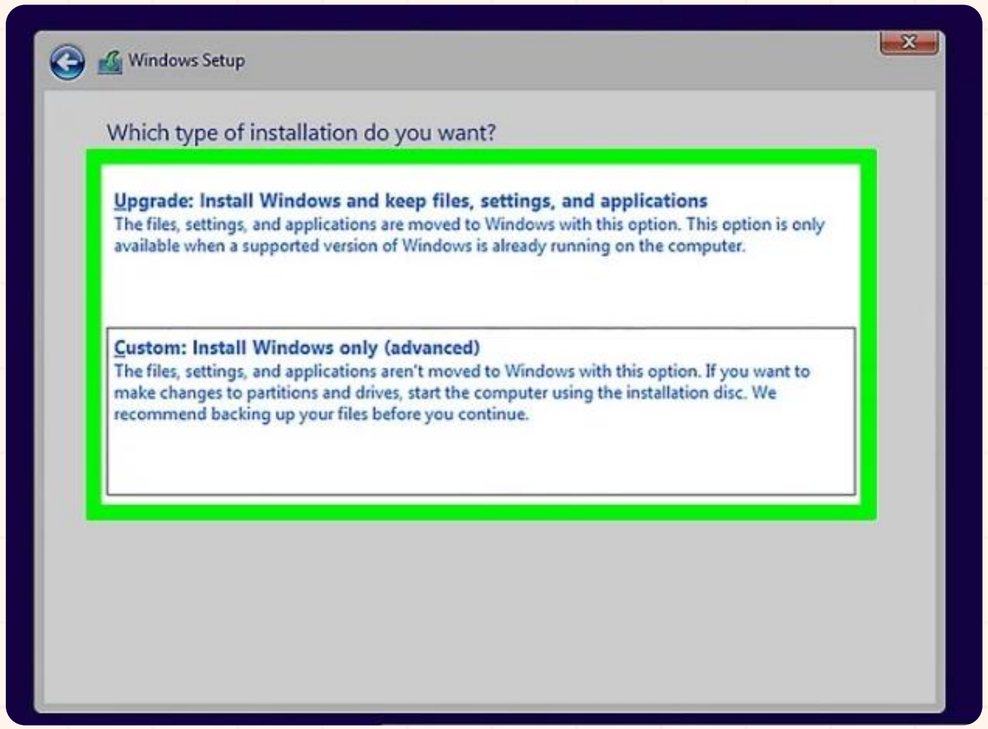
- You can read the license terms in the window in the center. Click next when you are ready to continue.



STEP 9:

SELECT CUSTOM OR UPGRADE OPTIONS

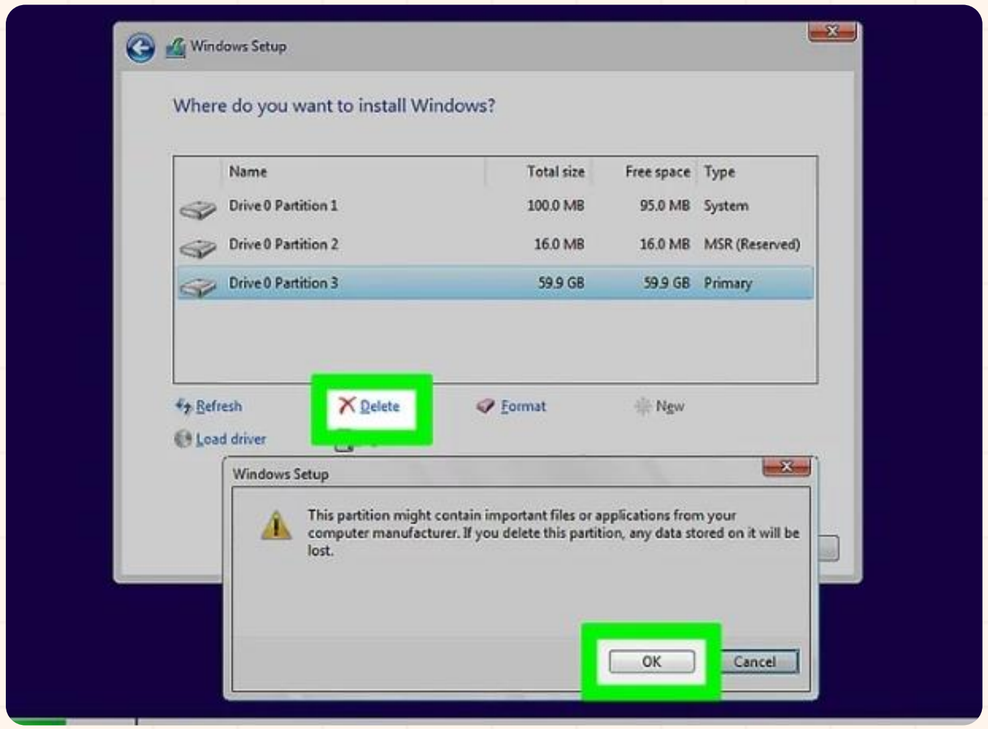
- If your computer already has a Windows installation on it, click Upgrade: Install Windows and keep files, settings, and applications. If your computer does not have a Windows installation or you need to reinstall Windows, click Custom: Install Windows only.



STEP 10:

select drive or partition

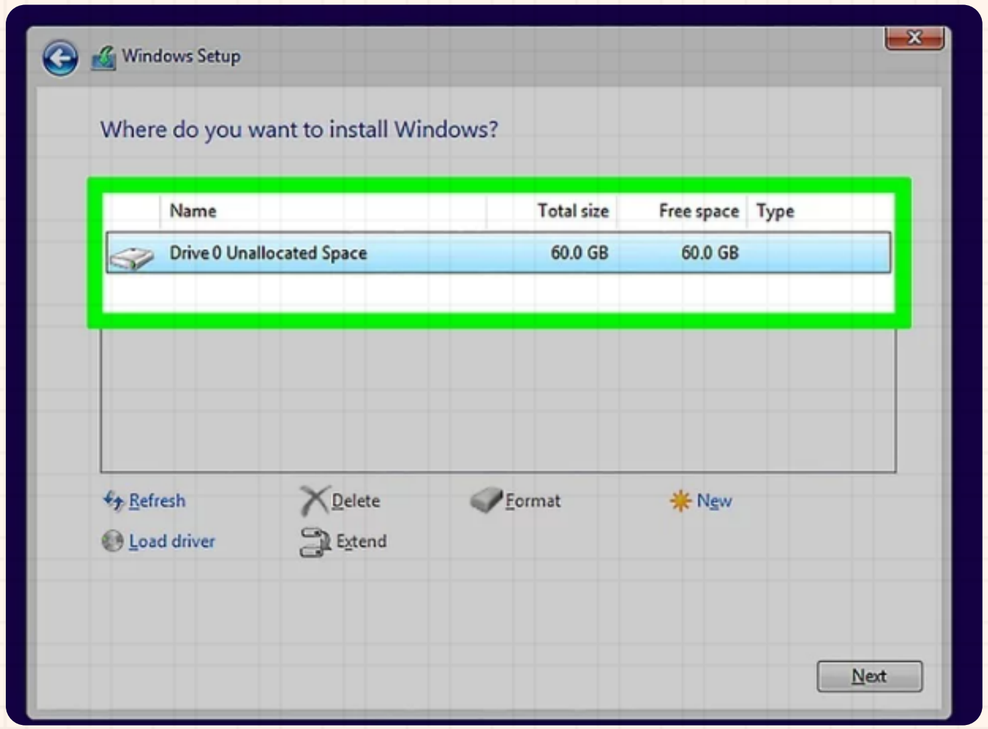
- If you have multiple hard drives or a hard drive with multiple partitions, select the partition you want to install Windows on and click Delete. Be aware that this will erase all data on the drive. Make sure you have backed up all data you want to keep before continuing. This will leave the drive with unallocated space.



STEP 11:

select a drive

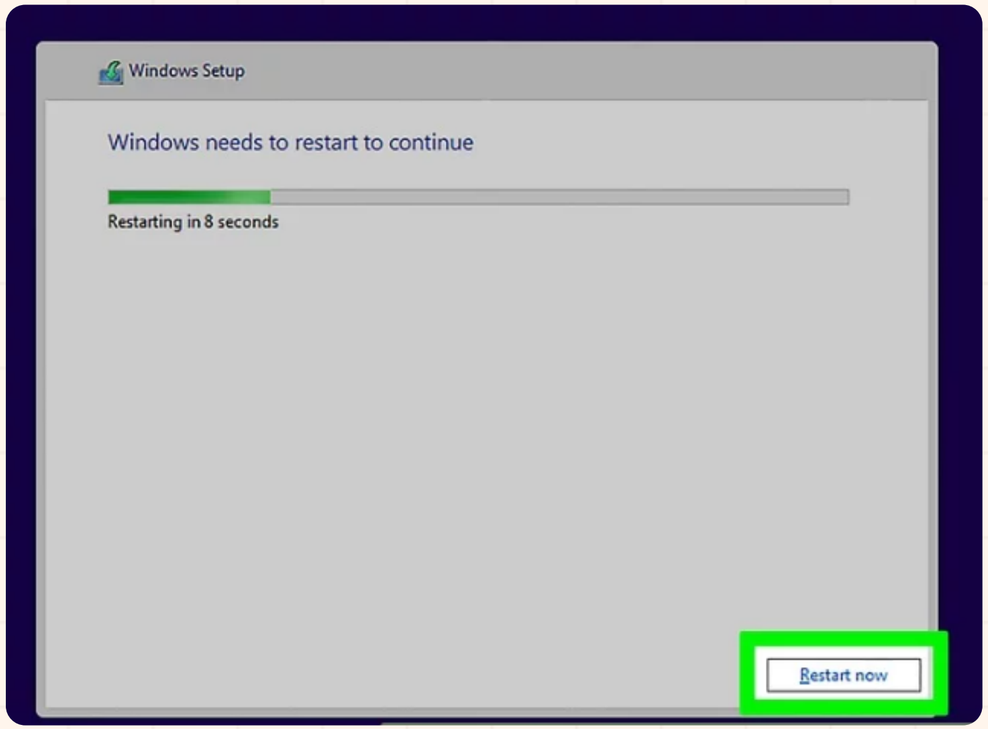
- This will start installing Windows on the drive you selected. The amount of time it takes will depend on the hardware of your computer.



STEP 12:

remove the usb

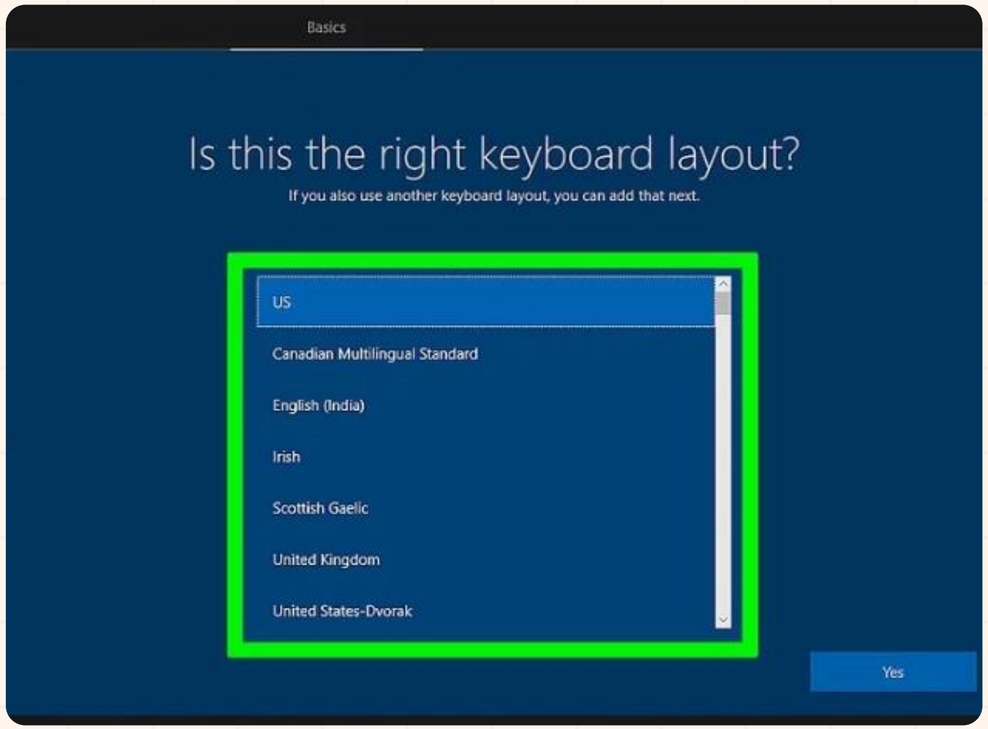
- Once Windows is finished installing, your computer will restart automatically. Remove the USB flash drive so that it does not try to boot from the Flash drive again. The first time Windows boots up, you will need to go through the Setup process.



STEP 13:

setting up windows

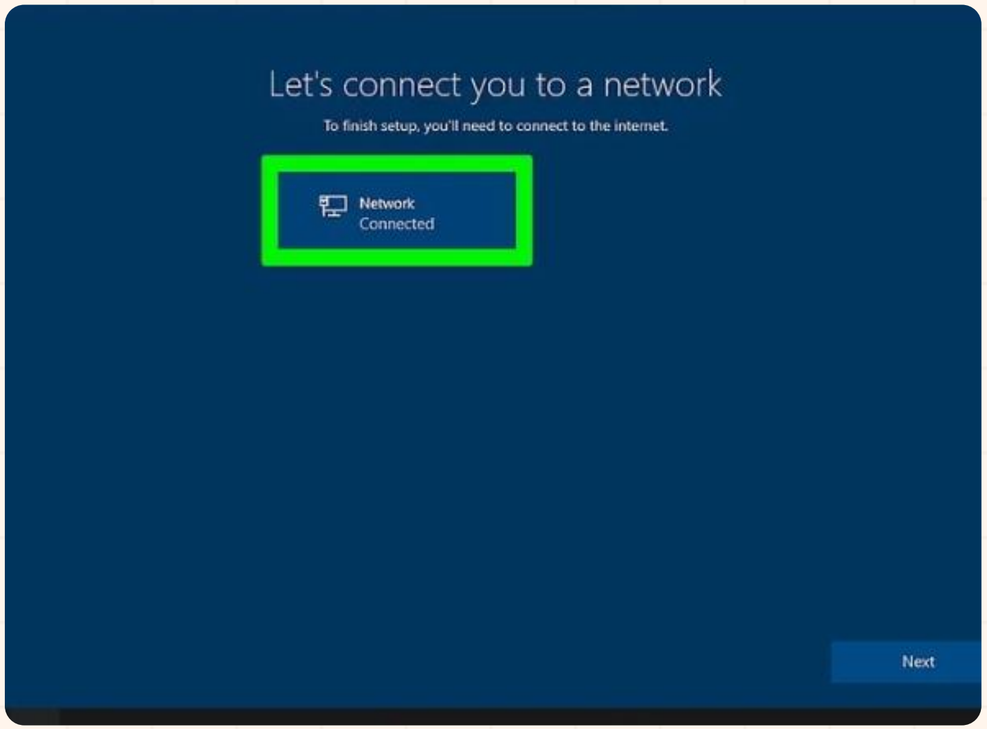
- When Windows restarts, it will ask you to verify your region and keyboard input. If these are correct, click Yes in the lower-right corner. Otherwise, select the correct region and keyboard input.



STEP 15:

connect to wireless network

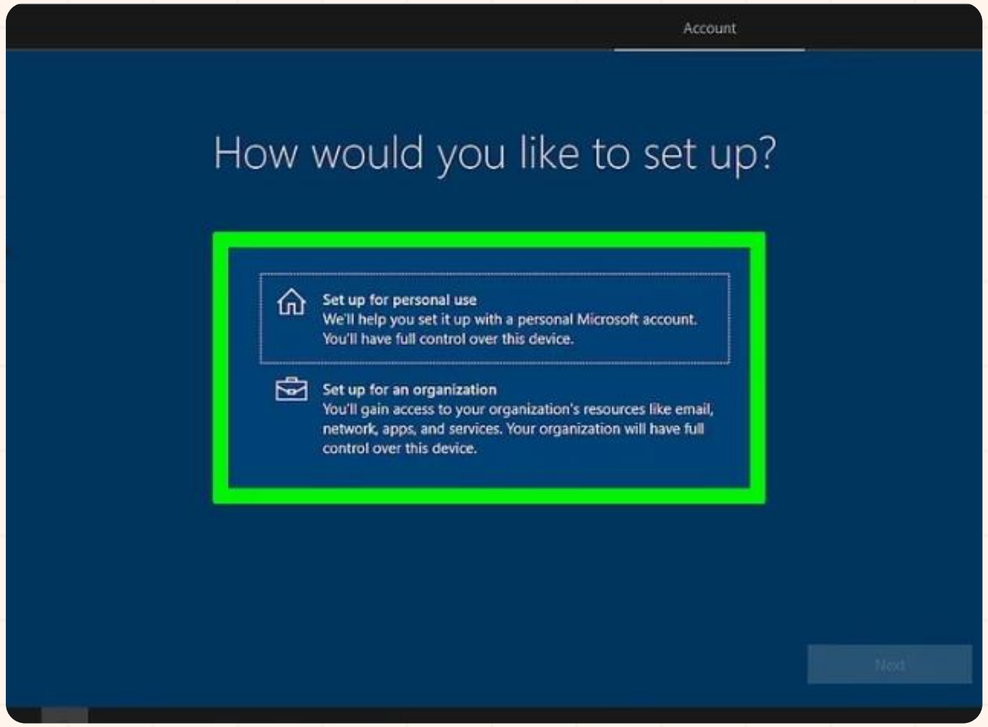
- If you have a wired internet connection, make sure you are connected to your modem via an Ethernet cable. Windows will automatically detect your wired connection.



STEP 16:

select a setup

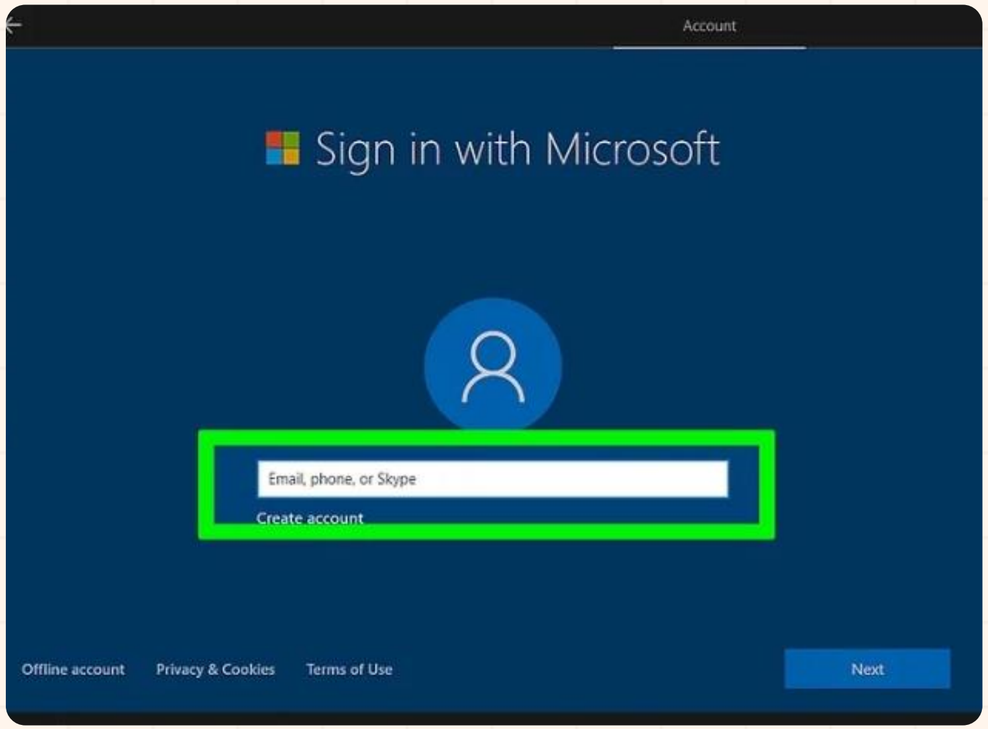
- If you are setting your computer for a business or organization, select Set up for an organization and click Next. You may need the aide of your IT department in order to set up Windows with a company email.



STEP 17:

sign in to windows

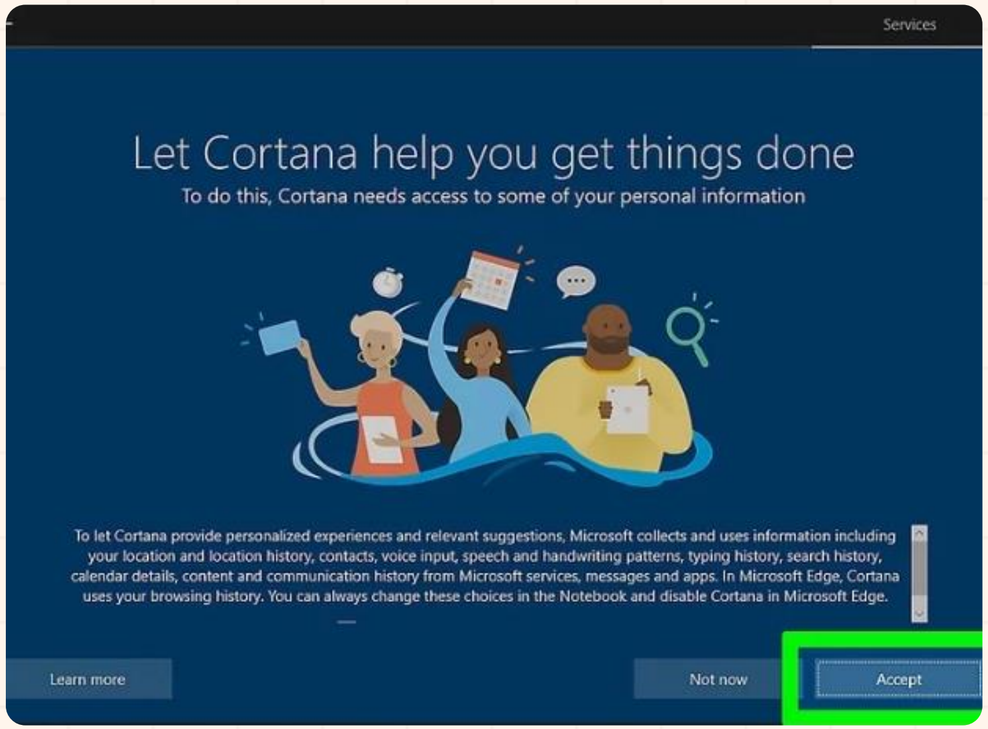
- If you do not have a Microsoft account, click Create account and follow the instructions to [create a new Microsoft account](#).
- Alternatively, you can click Offline account in the lower-right corner to create a new local user account for Windows that doesn't have a Microsoft account associated with it.



STEP 18:

accept cortana

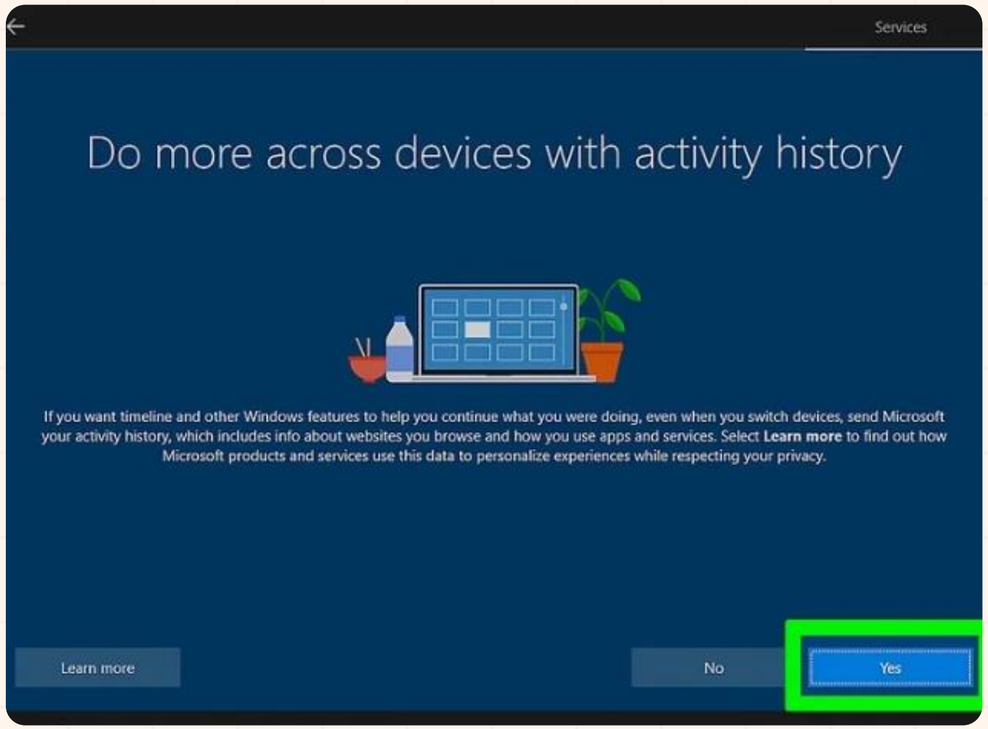
- Cortana is the built-in voice assistant for Windows. If you want to set up Cortana, click Accept and follow the instructions. If you don't want to do it right now, click Decline.



STEP 19:

click yes and follow the instructions

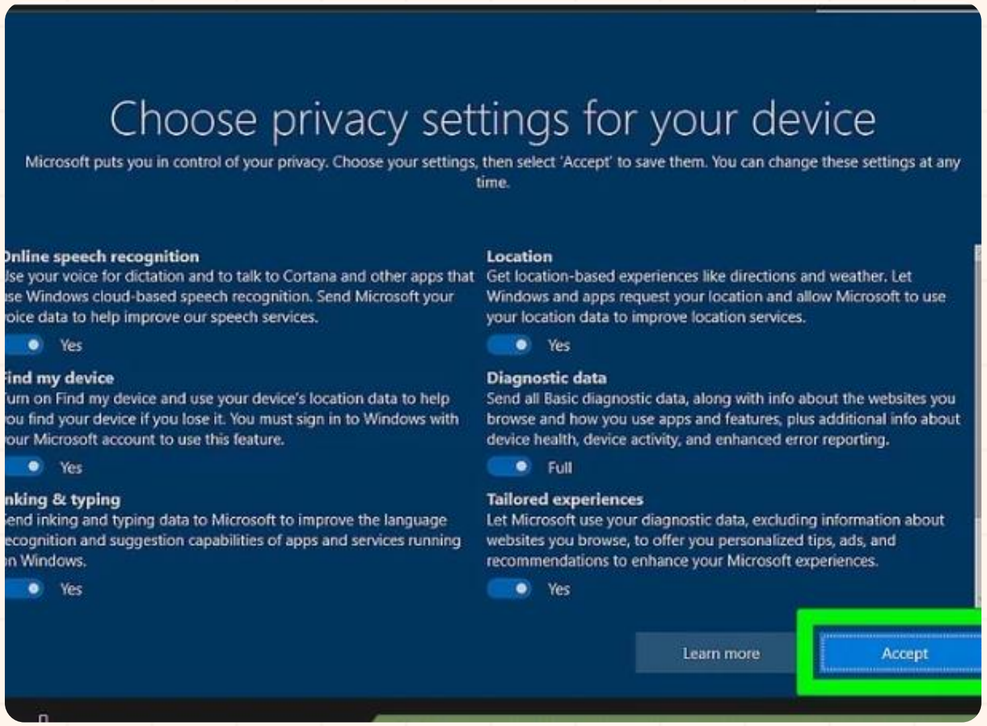
- This feature allows you to continue what you are doing on other Windows devices. If you want to set up this feature click Yes and follow the instructions. If you don't want to set this feature up, click No to move on to the next step.



STEP 20:

choose privacy

- By default, Windows turn on GPS location tracking, Speech recognition, diagnostics, find my device, inking and typing to help Microsoft improve language recognition, and relevant ads. Click the toggle switch to turn off any of these features you don't want to use. Then click Accept. Windows will do a little more set up and then open your desktop.



ACTIVITY 1

MAKE A STEP-BY-STEP GUIDE TO HARD DRIVE FORMATTING AND OPERATING SYSTEM INSTALLATION

ACTIVITY 2

MULTIPLE CHOICE

1. WHAT DOES FORMATTING A HARD DRIVE DO?

A. Hides all files withOUT DELETING ANYTHING
B. PREPARES THE STORAGE SO IT CAN STORE DATA AND AN OPERATING SYSTEM
C. INCREASES INTERNET SPEED
D. CHANGES THE COLOR OF THE HARD DRIVE

2. BEFORE FORMATTING A DRIVE, WHAT SHOULD YOU ALWAYS DO?

A. TURN OFF THE COMPUTER
B. BACK UP IMPORTANT DATA
C. INSTALL ANTIVIRUS SOFTWARE
D. DEFRAGMENT THE DRIVE

3. WHICH OF THESE IS A COMMON FILE SYSTEM USED BY WINDOWS?

A. NTFS
B. MP3
C. PDF
D. DOC

4. WHERE DO YOU USUALLY START WHEN INSTALLING AN OPERATING SYSTEM?

A. Inside File Explorer
B. FrOM INSTALLATION MEDIA (USB/DVD)
C. FROM TASK MANAGER
D. FROM CONTROL PANEL

5. YOU CANNOT FORMAT THE SYSTEM DRIVE (LIKE C:) WHILE WINDOWS IS RUNNING. WHY?

A. IT'S TOO SLOW
B. THE OS CANNOT MODIFY ITSELF WHILE RUNNING
C. IT INCREASES BRIGHTNESS
D. IT DELETES DRIVERS

6. WHAT HAPPENS TO DATA ON A HARD DRIVE WHEN YOU FORMAT IT?

A. IT STAYS THE SAME
B. IT GETS DELETED
C. IT BECOMES HIDDEN
D. IT COMPRESSES AUTOMATICALLY

7. WHY CHOOSE A SPECIFIC file system (like NTFS) whEN INSTALLING AN OS?

A. IT AFFECTS HOW DATA IS STORED AND MANAGED
B. IT CHANGES DESKTOP WALLPAPERS
C. IT SLOWS DOWN THE COMPUTER
D. IT INCREASES BATTERY LIFE

8. WHAT IS A QUICK FORMAT?

A. A FORMAT THAT DELETES FILES FASTER
B. A FORMAT THAT SILENTLY INSTALLS AN OS
C. A FORMAT THAT TAKES HOURS
D. A FORMAT DONE ONLY ON USB DRIVES

9. TO INSTALL AN OS, WHAT DO YOU NEED TO SELECT DURING SETUP?

A. THE CORRECT LANGUAGE AND KEYBOARD
B. THE HARD DRIVE OR PARTITION TO INSTALL ONTO
C. THE FILE SYSTEM TYPE
D. ALL OF THE ABOVE

10. WHICH OF THESE IS A REASON FOR formatting a drive bEFORE INSTALLING AN OS?

A. TO IMPROVE AUDIO QUALITY
B. TO ERASE OLD DATA AND PREPARE FOR NEW OS
C. TO CHANGE THE CASE COLOR
D. TO TIDY DESKTOP ICONS

11. WHAT DO YOU USE TO MAKE A USB DRIVE THAT CAN INSTALL AN OS?

A. A PHOTO EDITOR
B. A BOOTABLE USB CREATION TOOL
C. A MUSIC PLAYER
D. A WEB BROWSER

12. WHAT DOES “PARTITIONING” A DRIVE DO?

A. SAVES ENERGY
B. DIVIDES THE DRIVE INTO SEPARATE AREAS
C. MAKES YOUR SCREEN BRIGHTER
D. INCREASES FAN SPEED

13. IF YOU FORMAT THE WRONG drive by mistake, what happENS?

A. NOTHING
B. YOU LOSE THE DATA ON THAT DRIVE
C. THE COMPUTER EXPLODES
D. THE OS UPDATES AUTOMATICALLY

14. WHICH STORAGE DEVICE CAN BE FORMATTED AND THEN USED TO INSTALL AN OS?

A. CALCULATOR
B. USB FLASH DRIVE
C. PRINTER CABLE
D. HEADPHONES

15. WHICH STEP IS PART OF THE OS INSTALLATION AFTER FORMATTING?

A. INSTALLING THE OS FILES
B. CHANGING PRINTER SETTINGS
C. TURNING OFF WI-FI
D. DRAWING A LOGO

16. WHEN FORMATTING before OS INSTALLATION, THE PROCESS CLEARS WHAT FROM THE DRIVE?

A. ONLY TEMPORARY FILES
B. ALL EXISTING STRUCTURES AND DATA
C. THE WALLPAPER ONLY
D. NONE OF THE ABOVE

17. WHAT TOOL IS MOST COMMONLY USED TO CREATE A BOOTABLE USB FOR OS INSTALLATION?

A. A MUSIC APP
B. MEDIA CREATION SOFTWARE
C. VIDEO EDITOR
D. SPREADSHEET

18. AFTER FORMATTING, IF THE DRIVE DOESN'T HAVE AN OS, WHAT MUST YOU DO?

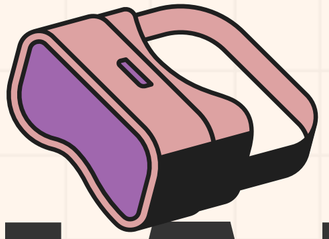
A. WAIT FOR 24 HOURS
B. INSTALL AN OPERATING SYSTEM
C. RUN ANTIVIRUS
D. UNPLUG THE KEYBOARD

19. WHICH OF THESE MAY BE erased by FORMATTING?

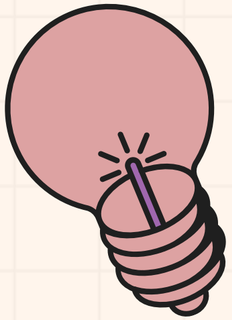
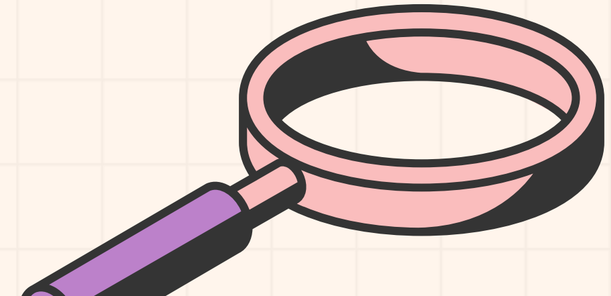
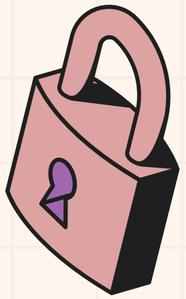
A. DESKTOP BACKGROUND
B. ALL PERSONAL FILES
C. KEYBOARD LAYOUT
D. SCREENSHOT SETTINGS

20. WHY DO WE FORMAT A DRIVE BEFORE INSTALLING AN OS?

A. TO PREPARE IT FOR THE NEW FILE SYSTEM
B. TO PLAY GAMES
C. TO WATCH VIDEOS
D. TO PRINT DOCUMENTS



THANK YOU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